

114 學年度第二學期 離散數學 HW08

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1 題目

1.1 作業繳交說明

- 從 overleaf 下載 hw08.zip
- 從 overleaf 下載 hw08.pdf
- 建立一個新檔案夾，命名如下
- 離散數學_114590xxx_姓名_HW08
- 放入 hw08.zip
- 放入 hw08.pdf
- 將此檔案夾，壓縮成 zip 檔，檔名如下
- 離散數學_114590xxx_姓名_HW08.zip
- 將此壓縮檔，上傳到北科 i 學園 PLUS
- Do not share your homework with any living creatures.
- 作業遲交以零分計。

1.2 題號

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2 作答

2.1 page 552, chapter 8.2 Exercise 26

What is the general form of the particular solution guaranteed to exist by Theorem 6 of the linear non-homogeneous recurrence relation $a_n = 6a_{n-1} - 12a_{n-2} + 8a_{n-3} + F(n)$ if

- $F(n) = n^2$?
- $F(n) = 2^n$?
- $F(n) = n2^n$?
- $F(n) = (-2)^n$?
- $F(n) = n^2 2^n$?
- $F(n) = n^3(-2)^n$?
- $F(n) = 3$?

We need to use Theorem 6, and so we need to find the roots of the characteristic polynomial of the associated homogeneous recurrence relation. The characteristic equation is $r^3 - 6r^2 + 12r - 8 = 0$, use $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$, $r = 2$ is the only root, and it has multiplicity 3.

- In the notation of Theorem 6, $s = 1$ here. Since 1 is not a root of the characteristic polynomial of the associated homogeneous recurrence relation, Theorem 6 tells us that the particular solution will be of the form $p_2 n^2 + p_1 n + p_0$.
- Since 2 is a root with multiplicity 3 of the characteristic polynomial of the associated homogeneous recurrence relation, Theorem 6 tells us that the particular solution will be of the form $p_0 n^3 2^n$.
- Since 2 is a root with multiplicity 3 of the characteristic polynomial of the associated homogeneous recurrence relation, Theorem 6 tells us that the particular solution will be of the form $2^n(p_1 n + p_0)n^3$.
- Since -2 is not a root of the characteristic polynomial of the associated homogeneous recurrence relation, Theorem 6 tells us that the particular solution will be of the form $p_0(-2)^n$.
- Since 2 is a root with multiplicity 3 of the characteristic polynomial of the associated homogeneous recurrence relation, Theorem 6 tells us that the particular solution will be of the form $(p_2 n^5 + p_1 n^4 + p_0 n^3)2^n$.
- Since -2 is not a root of the characteristic polynomial of the associated homogeneous recurrence relation, Theorem 6 tells us that the particular solution will be of the form $(p_3 n^3 + p_2 n^2 + p_1 n + p_0)(-2)^n$.
- Since 1 is not a root of the characteristic polynomial of the associated homogeneous recurrence relation, Theorem 6 tells us that the particular solution will be of the form p_0 . In the notation of Theorem 6, $s = 1$ here.

2.2 page 552, chapter 8.2 Exercise 28

- Find all solutions of the recurrence relation $a_n = 2a_{n-1} + 2n^2$.
- Find the solution of the recurrence relation in part (a) with initial condition $a_1 = 4$.

可以參考課本第 548 頁的 Example 10 and 11

- The associated homogeneous recurrence relation is $a_n = 2a_{n-1}$. We easily solve it to obtain $a_n^{(h)} = \alpha 2^n$. Next we need a particular solution to the given recurrence relation. By Theorem 6 we want to look for a function of the form $a_n = p_2 n^2 + p_1 n + p_0$. (Note that $s = 1$ here, and 1 is not a root of the characteristic polynomial.) We plug this into our recurrence relation and obtain $p_2 n^2 + p_1 n + p_0 = 2(p_2(n-1)^2 + p_1(n-1) + p_0) + 2n^2$. We rewrite this by grouping terms with equal powers of n , obtaining $(-p_2 - 2)n^2 + (4p_2 - p_1)n + (-2p_2 + 2p_1 - p_0) = 0$. In order for this equation to be true for all n , we must have $p_2 = -2$, $4p_2 = p_1$, and $-2p_2 + 2p_1 - p_0 = 0$. This tells us that $p_1 = -8$ and $p_0 = -12$. Therefore the particular solution we seek is $a_n^{(p)} = -2n^2 - 8n - 12$. So the general

solution is the sum of the homogeneous solution and this particular solution, namely $a_n = \alpha 2^n - 2n^2 - 8n - 12$.

- (b) We plug the initial condition into our solution from part(a) to obtain $4 = a_1 = 2\alpha - 2 - 8 - 12$. This tells us that $\alpha = 13$. So the solution is $a_n = 13 \cdot 2^n - 2n^2 - 8n - 12$.

2.3 page 552, chapter 8.2 Exercise 30

- (a) Find all solutions of the recurrence relation $a_n = -5a_{n-1} - 6a_{n-2} + 42 \cdot 4^n$.
- (b) Find the solution of this recurrence relation with $a_1 = 56$ and $a_2 = 278$.
- (a) The associated homogeneous recurrence relation is $a_n = -5a_{n-1} - 6a_{n-2}$. To solve it we find the characteristic equation $r^2 + 5r + 6 = 0$, find that $r = -2$ and $r = -3$ are its solutions, and therefore obtain the homogeneous solution $a_n^{(h)} = \alpha(-2)^n + \beta(-3)^n$. Next we need a particular solution to the given recurrence relation. By Theorem 6 we want to look for a function of the form $a_n = c \cdot 4^n$. We plug this into our recurrence relation and obtain $c \cdot 4^n = -5c \cdot 4^{n-1} - 6c \cdot 4^{n-2} + 42 \cdot 4^n$. We divide through by 4^{n-2} , obtaining $16c = -20c - 6c + 42 \cdot 16$, whence with a little simple algebra $c = 16$. Therefore the particular solution we seek is $a_n^{(p)} = 16 \cdot 4^n = 4^{n+2}$. So the general solution is the sum of the homogeneous solution and this particular solution, namely $a_n = \alpha(-2)^n + \beta(-3)^n + 4^{n+2}$.
- (b) We plug the initial conditions into our solution from part (a) to obtain $56 = a_1 = -2\alpha - 3\beta + 64$ and $278 = a_2 = 4\alpha + 9\beta + 256$. A little algebra yields $\alpha = 1$ and $\beta = 2$. So the solution is $a_n = (-2)^n + 2(-3)^n + 4^{n+2}$.

2.4 page 552, chapter 8.2 Exercise 32

Find the solution of the recurrence relation $a_n = 2a_{n-1} + 3 \cdot 2^n$.

The associated homogeneous recurrence relation is $a_n = 2a_{n-1}$. We easily solve it to obtain $a_n^{(h)} = \alpha 2^n$. Next we need a particular solution to the given recurrence relation. By Theorem 6 we want to look for a function of the form $a_n = cn \cdot 2^n$. We plug this into our recurrence relation and obtain $cn \cdot 2^n = 2c(n-1)2^{n-1} + 3 \cdot 2^n$. We divide through by 2^{n-1} , obtaining $2cn = 2c(n-1) + 6$, whence with a little simple algebra $c = 3$. Therefore the particular solution we seek is $a_n^{(p)} = 3n \cdot 2^n$. So the general solution is the sum of the homogeneous solution and this particular solution, namely $a_n = \alpha 2^n + 3n \cdot 2^n$.

2.5 page 575, chapter 8.4 Exercise 4

Find a closed form for the generating function for each of these sequences. (Assume a general form for the terms of the sequence, using the most obvious choice of such a sequence.)

- (a) $-1, -1, -1, -1, -1, -1, 0, 0, 0, 0, 0, \dots$
- (b) $1, 3, 9, 27, 81, 243, 729, \dots$

- (c) $0, 0, 3, -3, 3, -3, 3, -3, \dots$
- (d) $1, 2, 1, 1, 1, 1, 1, 1, \dots$
- (e) $\binom{7}{0}, 2\binom{7}{1}, 2^2\binom{7}{2}, \dots, 2^7\binom{7}{7}, 0, 0, 0, 0, \dots$
- (f) $-3, 3, -3, 3, -3, 3, \dots$
- (g) $0, 1, -2, 4, -8, 16, -32, 64, \dots$
- (h) $1, 0, 1, 0, 1, 0, 1, 0, \dots$

We will use Table 1 in much of this solution.

- (a) Apparently all the terms are 0 except for the seven -1 's shown. Thus $f(x) = -1 - x - x^2 - x^3 - x^4 - x^5 - x^6$. This is already in closed form, but we can also write it more compactly as $f(x) = \frac{x^7-1}{1-x}$, making use of the identity from Example 2.
- (b) This sequence fits the pattern in Table 1 for $1/(1-ax)$ with $a = 3$. Therefore the generating function is $1/(1-3x)$.
- (c) We can factor out $3x^2$ and write the generating function as $\frac{3x^2}{1+x}$, again using the identity in Table 1.
- (d) Except for the extra x (the coefficient of x is 2 rather than 1), the generating function is just $1/(1-x)$. Therefore the answer is $x + (1/(1-x))$.
- (e) From Table 1, we see that the binomial theorem applies and we can write this as $(1+2x)^7$.
- (f) We can factor out -3 and write the generating function as $\frac{-3}{1+x}$, using the identity in Table 1.
- (g) We can factor out x and write the generating function as $\frac{x}{1+2x}$, using the sixth identity in Table 1 with $a = -2$.
- (h) From Table 1 we see that the generating function here is $\frac{1}{1-x^2}$.

2.6 page 576, chapter 8.4 Exercise 16

Use generating functions to find the number of ways to choose a dozen bagels from three varieties—egg, salty, and plain—if at least two bagels of each kind but no more than three salty bagels are chosen.

The factors in the generating function for choosing the egg and plain bagels are both $x^2 + x^3 + x^4 + \dots$. The factor for choosing the salty bagels is $x^2 + x^3$. Therefore the generating function for this problem is $(x^2 + x^3 + x^4 + \dots)^2(x^2 + x^3)$. We want to find the coefficient of x^{12} , since we want 12 bagels. This is equivalent to finding the coefficient of x^6 in $(1 + x + x^2 + \dots)^2(1 + x)$. This function is $(1+x)/(1-x)^2$, so we want the coefficient of x^6 in $1/(1-x)^2$, which is 7, plus the coefficient of x^5 in $1/(1-x)^2$, which is 6. Thus the answer is 13.

2.7 page 576, chapter 8.4 Exercise 18

Use generating functions to find the number of ways to select 14 balls from a jar containing 100 red balls, 100 blue balls, and 100 green balls so that no fewer than 3 and no more than 10 blue balls

are selected. Assume that the order in which the balls are drawn does not matter.

Without changing the answer, we can assume that the jar has an infinite number of balls of each color; this will make the algebra easier. For the red and green balls the generating function is $1 + x + x^2 + \dots$, but for the blue balls the generating function is $x^3 + x^4 + \dots + x^{10}$, so the generating function for the whole problem is $(1 + x + x^2 + \dots)^2(x^3 + x^4 + \dots + x^{10})$. We seek the coefficient of x^{14} . This is the same as the coefficient of x^{11} in

$$(1 + x + x^2 + \dots)^2(1 + x + \dots + x^7) = \frac{1 - x^8}{(1 - x)^3}$$

Since the coefficient of x^n in $1/(1 - x)^3$ is $C(n + 2, 2)$, and we have two contributing terms determined by the numerator, our answer is $C(13, 2) - C(5, 2) = 68$.

2.8 page 576, chapter 8.4 Exercise 24

- (a) What is the generating function for a_k , where a_k is the number of solutions of $x_1 + x_2 + x_3 + x_4 = k$ when x_1, x_2, x_3 , and x_4 are integers with $x_1 \geq 3, 1 \leq x_2 \leq 5, 0 \leq x_3 \leq 4$, and $x_4 \geq 1$?
- (b) Use your answer to part (a) to find a_7 .

- (a) The restriction on x_1 gives us the factor $x^3 + x^4 + x^5 + \dots$.
The restriction on x_2 gives us the factor $x + x^2 + x^3 + x^4 + x^5$.
The restriction on x_3 gives us the factor $1 + x + x^2 + x^3 + x^4$.
And the restriction on x_4 gives us the factor $x + x^2 + x^3 + \dots$.
Thus the answer is the product of these:

$$(x^3 + x^4 + x^5 + \dots)(x + x^2 + x^3 + x^4 + x^5)(1 + x + x^2 + x^3 + x^4)(x + x^2 + x^3 + \dots)$$

We can use algebra to rewrite this in closed form as $x^5(1 + x + x^2 + x^3 + x^4)^2/(1 - x)^2$.

- (b) We want the coefficient of x^7 in this series, which is the same as the coefficient of x^2 in the series for

$$\frac{(1 + x + x^2 + x^3 + x^4)^2}{(1 - x)^2} = \frac{1 + 2x + 3x^2 + \text{higher order terms}}{(1 - x)^2}$$

Since the coefficient of x^n in $1/(1 - x)^2$ is $n + 1$, our answer is $1 \cdot 3 + 2 \cdot 2 + 3 \cdot 1 = 10$.

2.9 page 577, chapter 8.4 Exercise 34

Use generating functions to solve the recurrence relation $a_k = 7a_{k-1}$ with the initial condition $a_0 = 5$.

This problem is like Example 16. First let $G(x) = \sum_{k=0}^{\infty} a_k x^k$. Then

$$xG(x) = \sum_{k=0}^{\infty} a_k x^{k+1} = \sum_{k=1}^{\infty} a_{k-1} x^k \text{ (by changing the name of the}$$

variable from k to $k + 1$). Thus

$$G(x) - 7xG(x) = \sum_{k=0}^{\infty} a_k x^k - \sum_{k=1}^{\infty} 7a_{k-1} x^k \quad (1)$$

$$= a_0 + \sum_{k=1}^{\infty} (a_k - 7a_{k-1}) x^k \quad (2)$$

$$= a_0 + 0 \quad (3)$$

$$= 5, \quad (4)$$

because of the given recurrence relation and initial condition. Thus $G(x)(1 - 7x) = 5$, so $G(x) = \frac{5}{1-7x}$. From Table 1 we know then that $a_k = 5 \cdot 7^k$.

2.10 page 577, chapter 8.4 Exercise 36

Use generating functions to solve the recurrence relation $a_k = 3a_{k-1} + 4^{k-1}$ with the initial condition $a_0 = 1$.

Let $G(x) = \sum_{k=0}^{\infty} a_k x^k$. Then $xG(x) = \sum_{k=0}^{\infty} a_k x^{k+1} = \sum_{k=1}^{\infty} a_{k-1} x^k$ (by changing the name of the variable from k to $k + 1$). Thus

$$G(x) - 3xG(x) = \sum_{k=0}^{\infty} a_k x^k - \sum_{k=1}^{\infty} 3a_{k-1} x^k \quad (5)$$

$$= a_0 + \sum_{k=1}^{\infty} (a_k - 3a_{k-1}) x^k \quad (6)$$

$$= 1 + \sum_{k=1}^{\infty} 4^{k-1} x^k \quad (7)$$

$$= 1 + x \sum_{k=1}^{\infty} 4^{k-1} x^{k-1} \quad (8)$$

$$= 1 + x \sum_{k=0}^{\infty} 4^k x^k \quad (9)$$

$$= 1 + x \cdot \frac{1}{1 - 4x} \quad (10)$$

$$= \frac{1 - 3x}{1 - 4x}. \quad (11)$$

Thus $G(x)(1 - 4x) = (1 - 3x)/(1 - 4x)$, so $G(x) = \frac{1}{1-4x}$. Therefore $a_k = 4^k$, from Table 1.